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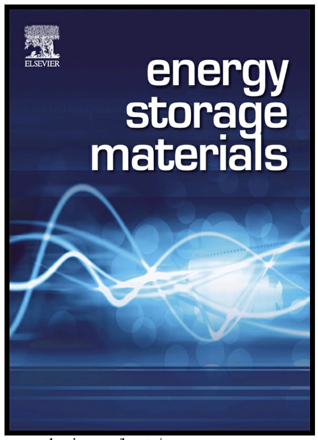


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# Synergistic electrocatalytic oxygen reduction reactions of $Pd/B_4C$ for ultra-stable Zn-air batteries

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**ABSTRACT:**

Highly active and durable electrocatalysts are desired to promote oxygen reduction reactions (ORR) for Zn-air batteries. Here we report a promising catalyst of Pd nanoparticles supported on boron carbide ($Pd/B_4C$) through a facile hydrothermal route. The $Pd/B_4C$ electrode not only shows the same onset potential of 0.96 V vs. RHE as the Pt/C electrode, but also an obviously larger limiting current density. More importantly, compared with noble metal electrodes, $Pd/B_4C$ shows higher power density (187 mW $cm^{-2}$) and much more durable cycle life (up to 1333 h) in Zn-air batteries. To gain insight into the enhanced ORR activity by $B_4C$ support, first-principles computations reveal the synergistic effect between $B_4C$ and Pd; $B_4C$ could significantly promote $O_2$ adsorption and splitting on $Pd/B_4C$ composites, while Pd particles on $B_4C$ have a major effect on the transformation of O* to $OH^-$. The

[1] Both authors contributed equally to this work.

computations are consistent with the experimental characterizations. This work opens up a new avenue for seeking novel and durable electrocatalysts for Zn-air batteries with $B_4C$ substrates, which are chemically and electrochemically stable in hash alkaline electrolytes during operation.

**Keywords**: Pd/$B_4C$, electrocatalysts, oxygen reduction reactions, Zn-air batteries

## 1. Introduction

To date, Zn-air batteries have attracted great attention due to their high theoretical energy density, which is much higher than that of lithium-ion batteries [1,2]. However, the sluggish kinetics related to charge transfer during oxygen reduction reactions (ORR) on the air cathode is the main bottleneck for metal-air batteries, leading to poor cycling stability and low power density [3-6]. To overcome this problem, highly active catalysts, including various carbon materials [7-10], metals/alloys [11-13], metal oxides [14-16], and metal carbides/nitrides [17,18], have been screened to catalyze the sluggish ORR. Despite much progress, most investigated catalysts still suffer from insufficient activity and poor stability under harsh alkaline conditions. Therefore, the exploration of high-performance and durable electrocatalysts is still urgent for ORR.

Pt and Pt-based alloys are widely recognized as the most active ORR catalysts in both acidic and alkaline electrolytes. Nevertheless, in view of the rarity and high cost of Pt, it is still imperative to find out appropriate cheaper alternatives with comparable catalytic activity and stability. As another Pt-group metal, Pd has been recognized as a promising cheaper alternative due to its corrosion-resistant character and ORR catalytic activity [19,20]. Unfortunately, Pd is still unacceptable for large-scale commercial applications, compared with nonprecious metal catalysts, and also, the ORR catalytic activity of conventional Pd catalysts is not high enough to replace Pt. As a consequence, effective substrates are highly desired to homogeneously disperse Pd particles while enhancing the catalytic activity synergistically.

Supported electro-catalysis is a widely accepted strategy to decrease the use of active species and simultaneously achieve high activity. In this case, the interaction between support materials and active species is critical for catalysts to generate high catalytic activity. Nitrogen-rich carbon materials are most commonly used as substrates owing to their high electrical conductivity and rich active sites [21-24]. However, N-rich carbon-based materials are susceptible to corrosive conditions and could be easily corroded by repetitively discharge and charge in harsh alkaline electrolytes [25,26]. Also with good electrical conductivity, particularly high chemical (no reactions with acidic and alkaline solutions) and electrochemical stability during operation, boron carbide ($B_4C$) has been attracting much attention as the catalyst support in electrochemical energy storage devices [27,28]. Using $B_4C$ as the support could enhance the catalytic activity through synergistic effects. Grubb and McKee [29] reported that compared with pure Pt, Pt on $B_4C$ showed enhanced activity in phosphoric acid fuel cells. Mu et al. [27] also investigated commercial $B_4C$ supported Pt in methanol fuel cells, which showed higher catalytic activity than conventional Pt/C catalysts. To the best of our knowledge, there has been no explicit report on the use of $B_4C$ as a catalyst support for relatively low-cost Pd in catalyzing ORR.

Here, we for the first time used $B_4C$ as the support to prepare Pd/$B_4C$ electrocatalysts and clarify the synergistic effect through the combination of experiments and computations. The as-prepared Pd/$B_4C$ exhibited significantly enhanced catalytic activity for ORR, as well as ultralong cycling durability in harsh alkaline electrolytes of Zn-air batteries. Both density functional theory (DFT) computations and experimental analyses reveal that the catalytic activity of Pd is obviously promoted by the introduction of $B_4C$, which could greatly improve the adsorption of $O_2$ and accelerate its splitting, thus boosting ORR kinetics.

## 2. Experimental

### *2.1 Materials Preparation*

All the reagents were analytical grade without further purification. Pd/$B_4C$ was prepared via a hydrothermal route. 100 mg $B_4C$ and 30 mg $PdCl_2$ were mixed with 35 mL ethylene glycol (EG) through intense agitation for 0.5 h, and then 0.9 g urea was added in the mixture and continually agitated for another 0.5 h. Then, we ultrasonically treated the mixture for 1 h in order to make it homogeneously dispersed, and the mixed solution was transferred into a 50 mL Teflon-lined stainless steel autoclave and kept at 180 ℃ for 1 h. After cooling, the precipitate was washed with distilled water and ethanol separately for three times, and dried at 60 ℃ in a vacuum oven. We also prepared Pd nanoparticles supported on Ketjen black (Pd/KB) as the control group. The preparation was the same as that of Pd/$B_4C$, except the substitution of commercial KB for $B_4C$.

*2.2 Materials characterizations*

X-ray diffraction (XRD) patterns were recorded on Rigaku MineFlex 600 with Cu Kα radiation. Elementary analyzer (vario EL CUBE, elementar) and inductively coupled plasma-atomic emission spectrometry (ICP-AES) were used to determine the content of different elements. Transmission electron microscope (TEM) and X-ray energy-dispersive spectrum (EDS) elemental mapping were performed on a FEI Tecnai G2F-20 field emission TEM. Aberration-corrected scanning transmission electron microscope (STEM, Titan Cubed Themis G2300) was also used to characterize the materials. $N_2$ adsorption/desorption isotherms were obtained by using Micromeritics ASAP 2460 Surface Area and Pore Size Analyzer. X-ray photoelectron spectroscopy (XPS, Axis Ultra DLD, Kratos Analytical) was also used to characterize the oxygen species on materials.

*2.3 Electrochemical measurements*

The electrochemical performances were measured by using a standard three-electrode electrochemical cell filled with different electrolytes at room temperature. To prepare the working electrode, the catalyst (5.0 mg) and 50 uL of 5 wt% Nafion solution were dispersed in 500 uL of distilled water and 450 uL of ethanol. The acquired mixture was sonicated to form a homogeneous suspension. Then the catalyst ink (5 uL) was pipetted on a commercial glassy carbon (GC) electrode (AFE5TOSOGC, 5 mm in diameter, 0.196 $cm^2$, Pine Research Instrumentation), and the sample was then dried at room temperature. Electrochemical tests were conducted with a CHI760E electrochemical working station. A glass carbon rotating disk electrode (RDE) covered with active materials was used as the working electrode while a Pt wire and Ag/AgCl electrode (saturated with KCl) were used as the counter and reference electrode, respectively. All potentials in this work were normalized to RHE according to the Nernst equation ($E_{RHE}$=$E_{Ag/AgCl}$+0.198+0.059×pH). During ORR tests, $O_2$ was flowed into the electrolyte for 30 min prior to the test and maintained over the electrolyte during the whole measurement to ensure the continued saturation of $O_2$ in the electrolyte. Before polarization tests, the working electrode was cycled at a scan rate of 100 $mVs^{-1}$ until stable current density was acquired. Polarization data were measured by using linear sweep voltammetry (LSV) at a scan rate of 5 $mVs^{-1}$ and various rotating speeds from 400 to 2025 rpm. The Koutecky-Levich equation was used to determine the electron transfer number (n) for the ORR process:

$$\frac{1}{J} = \frac{1}{J_k} + \frac{1}{B\omega^{1/2}}$$

$$\mathrm{B} = 0.2\mathrm{nF}C_o(D_o)^{2/3}\upsilon^{-1/6}$$

where $J$ is the measured current density, $J_k$ is the kinetic current, $\omega$ is the electrode rotation rate in rpm, F is the Faraday constant (96485 C $mol^{-1}$), $C_o$ is the bulk concentration of $O_2$ ($1.2\times10^{-6}$ mol $cm^{-3}$), $D_o$ is the diffusion coefficient of $O_2$ ($1.2\times10^{-6}$ $cm^2$ $s^{-1}$), and ν is the kinetic viscosity of electrolyte (0.01 $cm^2$ $s^{-1}$). The constant 0.2 is adopted when the rotation speed is expressed in rpm. The peroxide yields ($H_2O^-$) and number of transferred electrons (n) are calculated with the equations below:

$$H_2O^-(\%) = 100\frac{2I_r/N}{I_d + I_r/N}$$

$$n = 4\frac{I_d}{I_d + I_r/N}$$

where $I_d$ is the disk current, $I_r$ is the ring current, and $N$ is the current collection efficiency of the Pt ring, and is determined to be 0.41.

*2.4 Zn-air battery assembly*

The rechargeable Zn-air battery was tested by using 6 M KOH and 0.2 M $Zn(Ac)_2$ as the electrolyte, with a polished zinc plate as the anode. The air cathode was prepared by loading the catalyst ink onto carbon cloth fibers with mass loading of 0.6 mg $cm^{-2}$ (1 $cm^2$) and dried at 60 ℃. When assembling a Zn-air battery, we used stainless steel mesh (SSM) as the current collector and gas diffusion layer (GDL) as the water-proof protect layer to permit oxygen into the inner battery.

*2.5 Computational Details*

First-principles computations on basis of DFT were performed with the plane-wave technique as implemented in the Vienna ab initio simulation package (VASP) [30]. The projector augmented wave (PAW) pseudopotential [31] was utilized and a 500 eV cutoff was adopted for the plane wave basis set. The generalized gradient approximation (GGA) involving the Perdew, Burke, and Ernzerhof (PBE) functional was used [32]. In order to accurately deal with weak interactions, DFT-D3 method with Becke-Jonson damping was applied [33,34]. K-point separation of less than 0.03 $Å^{-1}$ was employed. The deformation electron density was computed by the total charge density subtracting the sum of the charge density of the isolated parts. A vacuum space with at least 15 Å was inserted between the surface adsorption model and its periodically repeated images. Considering the experimental results, we built the models of Pd(111) surface and $B_4C$(104) and (003) surface. The lower two layers of Pd and four layers of $B_4C$ were fixed while other atoms were fully relaxed. Besides, the models of Pd/$B_4C$ composites were also built. Considering the lattice mismatch of Pd on $B_4C$, we selected $B_4C$(003) and Pd(111) surface, which could cause a lattice mismatch of ~3%. The $B_4C$ atoms were fixed and the Pd atoms were fully relaxed. In alkaline environment, ORR would occur over Pd/$B_4C$ systems in the following four-electron reaction path:

$$O_2 + H_2O\,(l) + e^- \xrightarrow{\Delta G_1} OOH^* + OH^- \qquad (1)$$

$$OOH^* + e^- \xrightarrow{\Delta G_2} O^* + OH^- \qquad (2)$$

$$O^* + H_2O\,(l) + e^- \xrightarrow{\Delta G_3} OH^* + OH^- \qquad (3)$$

$$OH^* + e^- \xrightarrow{\Delta G_4} * + OH^- \qquad (4)$$

where * represents the active site on the catalyst surface, (1) refers to liquid phase, and O*, OH* and OOH* indicate adsorbed intermediates. Due to the inaccuracy of DFT in estimating the cohesive energy of $O_2$, the free energy change of $O_2$ was obtained from the free energy change of the reaction $O_2+2H_2\rightarrow2H_2O$, which is consistent with the previous report [35].

In order to understand the reaction process, the free energy change of each step was calculated based on the following equation:

$$\Delta G = \Delta E + \Delta ZPE - T\Delta S + \Delta G_U + \Delta G_{pH} \qquad (5)$$

in which ΔE stands for the reaction energy obtained by DFT computations, ΔZPE is the change of zero point vibrational energy. For entropy change, only vibrational entropy contributions were considered for the adsorbed species (O*, OH* and OOH*) and for the solvent molecules, the total entropies were considered, which was consistent with the previous report [36]. For $H_2O$, gas $H_2O$ at 0.035 bar was applied as the reference state since gas $H_2O$ is in equilibrium with liquid water at 300 K at this pressure. $\Delta G_U$ can be calculated by $\Delta G_U = -eU$, where $e$ represents the charge transferred and U means the potential at the electrode. $\Delta G_{pH}$ can be expressed by:

$$\Delta G_{pH} = -k_B T \ln [H^+] \qquad (6)$$

where $k_B$ is the Boltzmann constant. According to the experiments, pH of 13 was applied in the computations.

**3. Results and discussion**

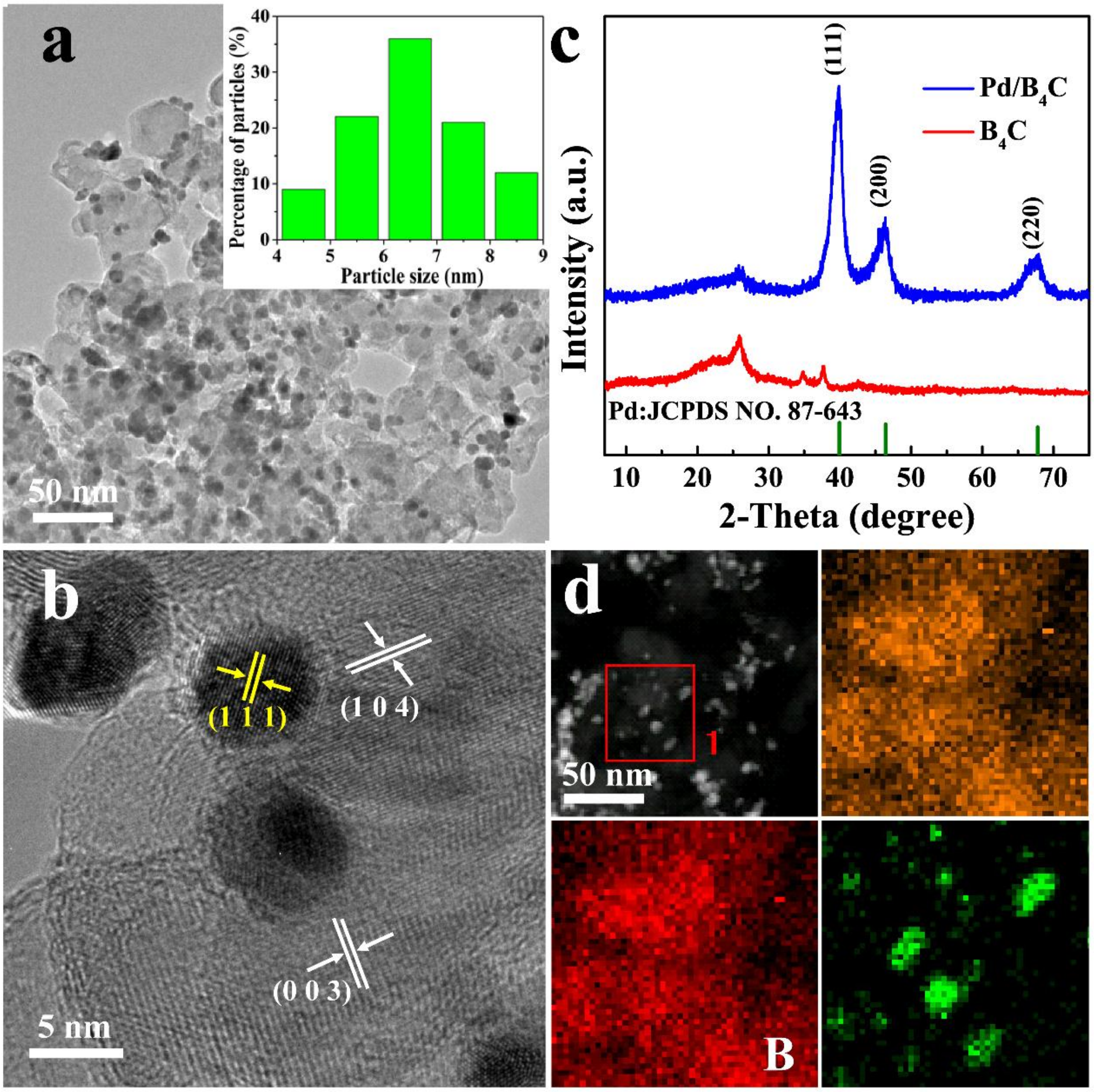


**Fig. 1** (a) TEM and (b) HRTEM images of $Pd/B_4C$ as well as (c) XRD patterns of $Pd/B_4C$ and $B_4C$. (d) EDS elemental mapping images of C, B and Pd in $Pd/B_4C$.

The morphology of $Pd/B_4C$ was observed through TEM. Fig. 1a shows the representative TEM image of Pd nanoparticles uniformly dispersed on the $B_4C$ support, which has a surface area of 68.7 $m^2$ $g^{-1}$ (Fig. S1). The average particle size of Pd is 6.5 nm. The exposed facet planes are (111) for Pd while (104) and (003) for $B_4C$ (Fig. 1b). XRD patterns (Fig. 1c) further confirm the existence of metallic Pd in the composite with diffraction peaks centered at 39.97°, 46.33° and 67.66°, which can be assigned to (111), (200) and (220) facets of Pd (JCPDS No. 87-643), respectively. The elemental distribution of Pd/$B_4C$ is also disclosed by EDS, showing the uniform distribution of C and B in the substrate and supported Pd nanoparticles (Fig. 1d). For comparison, Pd/KB was also prepared with an average size close to that of Pd/$B_4C$ (Fig. S2).

The electrocatalytic activity of the prepared catalysts towards ORR was evaluated on a rotating disk electrode (RDE) in $O_2$-saturated 0.1 M KOH solution. Cyclic voltammetry (CV) was first conducted in $N_2$ or $O_2$ saturated electrolytes at a scan rate of 50 mV $s^{-1}$ to investigate the ORR performance (Fig. 2a). In the $O_2$ saturated electrolyte, Pd/$B_4C$ exhibited an obvious oxygen reduction peak located at 0.71 V, and no redox peaks in the $N_2$ saturated electrolyte, confirming the ORR catalytic activity of Pd/$B_4C$. Fig. 2b shows LSV curves tested at 1600 rpm. Among the different catalysts, Pd/$B_4C$ exhibits the highest ORR activity, with almost the same onset potential (0.96 V vs. RHE) as the Pt/C (20 wt.%) electrode, along with an obviously higher limiting current density. Compared with the Pd/KB electrode, Pd/$B_4C$ shows a positive shift of about 20 mV of the ORR onset potential and obviously much higher limiting current density, indicating that the activity of Pd nanoparticles is higher on the $B_4C$ support than on the carbon substrate. The enhanced catalytic performance is most likely due to the synergistic effect of $B_4C$ and Pd, in which the electron deficiency of $B_4C$ makes it possible to promote the electron transfer from Pd to the $B_4C$ support, contributing to the formation of metal-support interaction. The very recent work by Colleen *et al*. [37] has also shown enhanced

ORR activity of Pt/$B_4C$. The mechanism would be further clarified through DFT computations and experimental characterizations in the following. Tafel slopes were then acquired from the polarization curves to investigate the difference about their ORR catalytic kinetics. As shown in Fig. 2c, Pd/$B_4C$ displays the lowest Tafel slope of 62 mV $dec^{-1}$, which is much lower than that of Pt/C (74 mV $dec^{-1}$), indicating the favourable ORR kinetics of the Pd/$B_4C$ electrode. The excellent ORR activity of Pd/$B_4C$ was also validated by the LSV test at different rotation speeds. As presented in Fig. 2d, Pd/$B_4C$ delivers much higher current density at different speeds than the benchmark Pt/C electrode (Fig. S3a). The Koutecky-Levich (K-L) plot (Fig. 2e) of the catalyst features good linearity and a consistent slope similar to that of Pt/C (Fig. S3b), from which the electron transfer number per oxygen molecule (n) for ORR was determined to be ~4.0, demonstrating a four-electron transfer process for ORR [38,39]. To further evaluate the ORR pathways for the Pd/$B_4C$ catalyst, we performed rotating ring-disk electrode (RRDE) measurements. The RRDE tests (Fig. 2f) discover that the Pd/$B_4C$ electrode can undergo the ORR process with an electron transfer number of 3.91-3.93 per oxygen molecule and a $HO_2^-$ yield generally below 6% in the potential region of 0-0.8 V in 0.1 M KOH solution, both of which are quite close to those from Pt/C (Fig. S4). In addition, the stability for ORR was also measured at the potential of 0.8 V under a rotating speed of 1600 rpm for Pd/$B_4C$ and Pt/C electrodes under the same condition for comparison. From Fig. S5, much improved stability could be obviously observed for Pd/$B_4C$, which outperforms the commercial Pt/C catalyst with the current retention of 96.5% vs. 83.4%. In addition, we further verified that Pd/$B_4C$

could also give excellent ORR activity in 0.5 M $H_2SO_4$ (Fig. S6), in which the Pd/$B_4C$ electrode shows obviously higher current density than the Pt/C electrode, though the ORR onset potential is slightly lower than that of the Pt/C electrode.

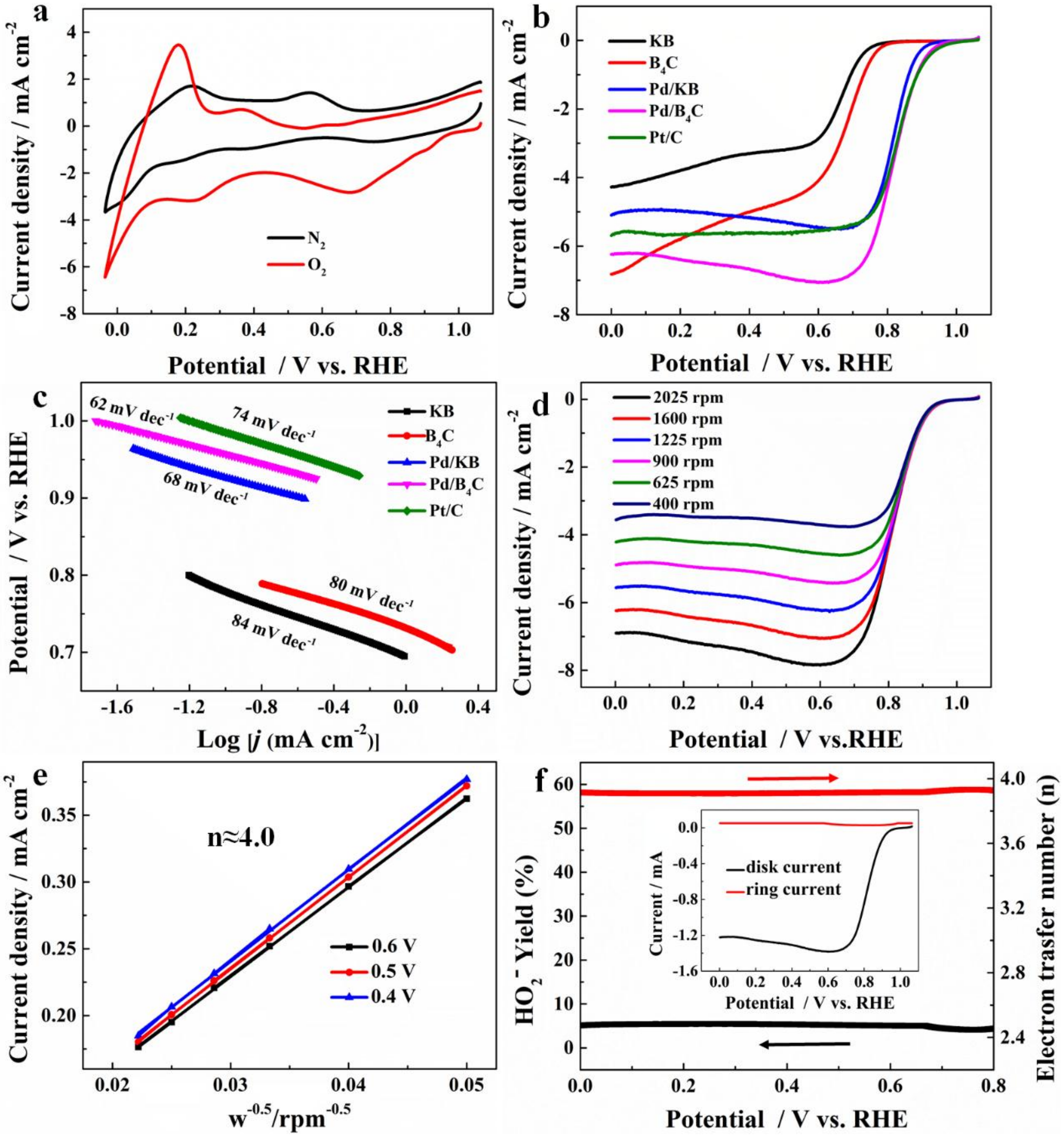


**Fig. 2** (a) CV curves for Pd/$B_4C$ in $N_2$ or $O_2$ saturated 0.1 M KOH solutions with a scan rate of 50 mV $s^{-1}$. (b) LSV curves of different catalysts for ORR in $O_2$-saturated 0.1 M KOH at 1600 rpm. (c) Tafel slopes derived from (b). (d) LSV curves of Pd/$B_4C$ for ORR at different rotating speeds. (e) K-L plots of Pd/$B_4C$ at different potentials

including the calculated electron transfer (n) per $O_2$. (f) Electron transfer number (n) and $HO_2^-$ yield derived from the RRDE tests, and the inset exhibits RRDE voltammograms of the Pd/$B_4C$ catalyst at 1600 rpm. The ring current is amplified by 5 for better resolution.

To gain insight into the reaction mechanisms of Pd/$B_4C$, we performed first-principles computations to investigate the catalytic activity of Pd and $B_4C$. The most exposed surfaces in experiments, Pd(111), $B_4C$(104) and (003) surfaces, were considered as the models for the investigations (Fig. S7). The elementary reaction pathways in alkaline environment are shown as reactions (1) to (4) (Experimental section). For ORR, the $O_2$ adsorption is one of the most important steps which could determine the ability to capture the reactant of the catalysts. Therefore, the adsorption for $O_2$ was firstly examined. For Pd(111) surface, several adsorption sites with end-on and bridge adsorption modes were investigated and the energetically most favourable configuration is shown in Fig. S8. A moderate adsorption energy of −0.83 eV was obtained, indicating that Pd(111) surface can effectively capture $O_2$ during ORR. Then, we computed the free energy of ORR at different electrode potentials in alkaline environment (pH = 13) as shown in Fig. S9. The last two steps of ORR are uphill when the potential is 0 V. However, when the potential reaches −0.62 V, the elementary reaction could become downhill which indicates that ORR can spontaneously occur. Moreover, according to the computations, during the transformation from step 3 to step 4, the formation of OH* is the rate-determining step in ORR. The overpotential $\eta$ is a significantly important indicator of ORR activity for a catalyst. Thus the minimum ORR overpotential was calculated based on 0.46 − (−0.62) = 1.08 V, comparable to that of Pt(111) [35], which could be observed experimentally as shown in Fig. 2b.

The catalytic activity of $B_4C$(104) and (003) surfaces was then investigated. Interestingly, different from the Pd(111) surface, after the adsorption of $O_2$ on $B_4C$ surface, the bond of $O_2$ was significantly elongated to 2.64 Å ((003) surface) and 2.76 Å ((104) surface) as shown in Fig. S10. The superior adsorption for $O_2$ ensures sufficient reactants, indicating an ultrahigh current density, which is consistent with our experimental results. Due to the great reactivity of $B_4C$ surface in oxygen splitting, only the free energy changes of reactions (3) and (4) ($\Delta G_3$ and $\Delta G_4$) were calculated to estimate the ORR activity of $B_4C$, and the results are shown in Table S1. However, the high $\Delta G$ suggests that when the potential reaches −2.00 V for (003) surface (−2.61 V for (104) surface), ORR would be downhill. These results indicate that oxygen reduction would launch on $B_4C$ at relatively low onset potential, which is consistent with the experimental results (Fig. 2b).

In addition, the ORR activity of Pd/$B_4C$ composites was also investigated computationally. Considering that Pd nanoparticles are highly dispersed on the $B_4C$ support as displayed in Fig. 1b, we built the models as shown in Fig. 3a-e. Also, the deformation electron density was calculated to investigate the interaction between Pd and $B_4C$ in Fig. 3f. The results indicate that due to the electron deficiency of $B_4C$, the electrons of Pd transfer to $B_4C$, contributing to stronger oxygen adsorption of Pd. Therefore, Pd/$B_4C$ composites exhibit a higher $O_2$ adsorption energy of −1.08 eV due to the existence of $B_4C$, indicating the stronger adsorption for $O_2$ and a larger current density at equilibrium than those of Pd without $B_4C$, which was also identified experimentally. The free energy of ORR was also calculated as shown in Fig. 3g. Due to the inclusion of $B_4C$, only an electrode potential of –0.38 V is needed to make the ORR downhill, which corresponds to a minimum ORR overpotential of 0.84 V, lower than that of Pd, indicating a better ORR activity and a higher onset potential, which are consistent with experiments. According to the experimental and computational results, we propose a possible mechanism for the ORR activity of Pd/$B_4C$ composites. Pd nanoparticles uniformly disperse on the $B_4C$ support, and $B_4C$ could effectively enhance the adsorption and splitting of $O_2$ on the Pd/$B_4C$ composites while Pd nanoparticles on $B_4C$ have a major effect in the transformation of O* to $OH^-$. The synergistic effect between Pd and $B_4C$ contributes to the improved ORR catalytic activity.

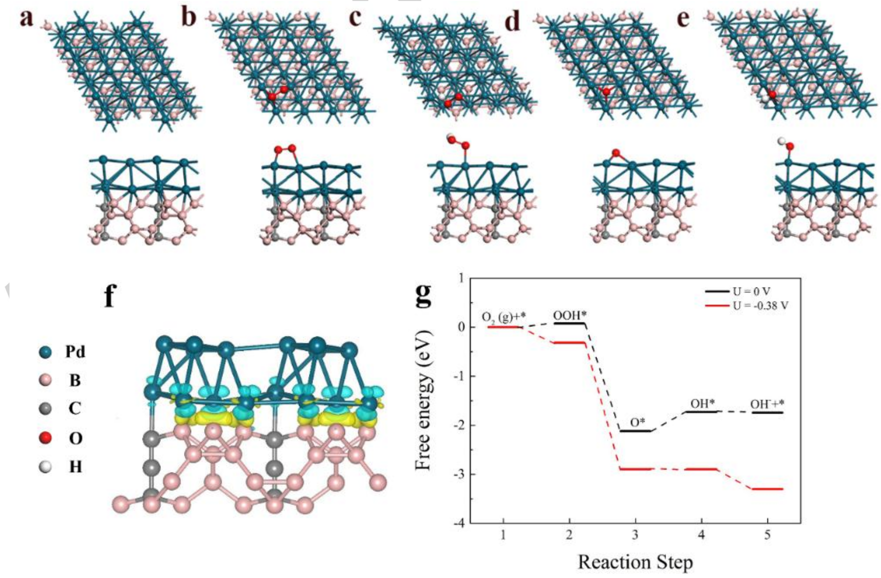


**Fig. 3** Top (up) and side (down) view of (a) the Pd/$B_4C$ composite and those after the adsorption of (b) $O_2$*, (c) OOH*, (d) O* and (e) OH*. (f) Deformation electron

density of $Pd/B_4C$ composites with an isosurface of 0.02 e $Å^{-3}$. The yellow and cyan regions represent the accumulation and depletion of the electron, respectively. (g) Free energy diagram for ORR on the $Pd/B_4C$ composite at different electrode potentials in alkaline media.

For comparison, the effect of KB on the ORR activity was also investigated in both experiments and computations. Here, a small carbon sheet was applied to simulate KB as shown in Fig. S11. Compared with $B_4C$ and Pd, the KB sheet exhibits extremely weak adsorption for $O_2$, implying the poor ORR activity and lower current density at the same potential as shown in Fig. 2b. To further evaluate the ORR activity, the free energy of ORR at different electrode potentials was also investigated in alkaline environment, as shown in Fig. S12. An electrode potential of −1.69 V is necessary to make the ORR elementary reaction downhill, indicating the weaker ORR activity and lower onset potential, which is consistent with the experimental results. Besides, compared with $B_4C$, the poor adsorption for $O_2$ of KB could also explain the weaker ORR activity and lower limiting current density of Pd/KB than that of $Pd/B_4C$.

In order to confirm the accuracy of the configuration describing the interaction between Pd and $B_4C$, electron energy loss spectroscopy (EELS) was conducted on an aberration-corrected STEM. The EELS was acquired from the point X to Y (Fig. S13), which represents the energy loss change from the inner part of the Pd nanoparticle to the interface of Pd and $B_4C$. We can easily observe that the peak at about 51 eV of Pd shifts to the higher direction, indicating the electron transfer from the Pd nanoparticle to $B_4C$ substrate at the interface.

To further verify the proposed mechanism from computations, XPS was conducted for the catalysts treated in 0.1 M KOH with saturated $O_2$. As shown in Fig. 4, the fitted XPS peaks for O1s centred at ~530.5 eV, 531.6 eV and 532.6 eV are attributed to oxygen in OH groups, adsorbed O and O in the multiplicity of physi- or chemisorbed water molecules on the surface of the composites [16,40,41], respectively. Besides these three different types of oxygen, the peak at 533.2 eV in Fig. 4a corresponds to B-O group existing in Pd/$B_4C$ [42,43]. Also, there is a broad peak at 533.6 eV in Fig. 4b that could be ascribed to C-O-C or C-O-OH groups in Pd/KB [44]. Despite these possible assignments, the peak at 531.6 eV of Pd/$B_4C$ catalyst occupies ~70% of the total area of XPS, which is much higher than that of Pd/KB (~31%). This indicates that the adsorbed O takes up a dominate proportion in the O1s spectra of Pd/$B_4C$, while only a small proportion in Pd/KB. The XPS results, which are in good agreement with computations, further account for the excellent catalytic activity of Pd/$B_4C$ composites.

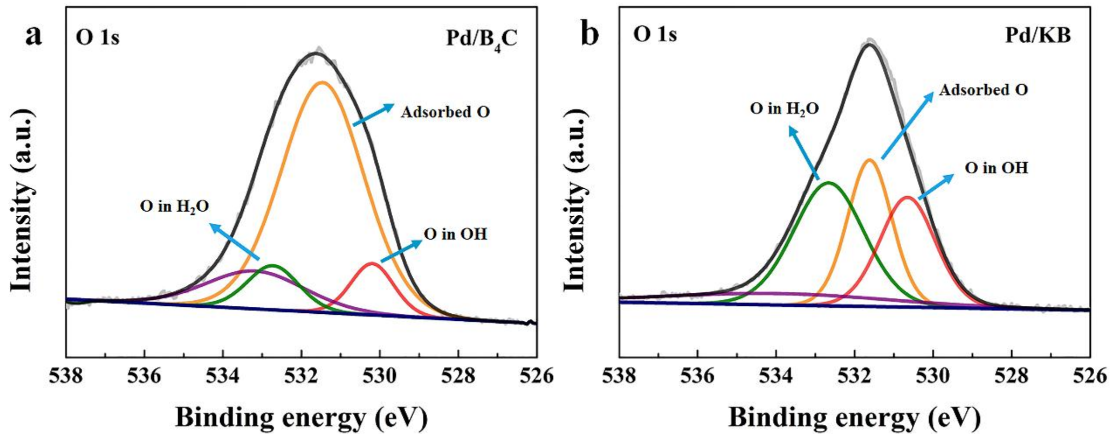


**Fig. 4** (a) High-resolution XPS of O1s for Pd/$B_4C$ and Pd/KB.

As the reverse reaction of ORR, OER is also imperative in the charge process of a rechargeable Zn-air battery. As shown in Fig. S14a, Pd/$B_4C$ shows a relatively low overpotential compared with other electrodes. The fast OER kinetic of Pd/$B_4C$ is further confirmed by its smaller Tafel slope (Fig. S14b). To elucidate the excellent OER activity, electrochemical impedance spectroscopy (EIS) was performed at 1.6 V vs. RHE. Fig. S15 exhibits a relatively smaller semicircle, revealing the decreased charge transfer resistance ($R_{ct}$) of Pd/$B_4C$. In addition, the electrochemical active surface area (ECSA) was also determined to elucidate the excellent oxygen catalytic activity. As revealed in Fig. S16, Pd/$B_4C$ exhibits the highest electrochemical double-layer capacitance ($C_{dl}$), which implies the existence of more abundant active surface area compared with Pt/C and $IrO_2$.

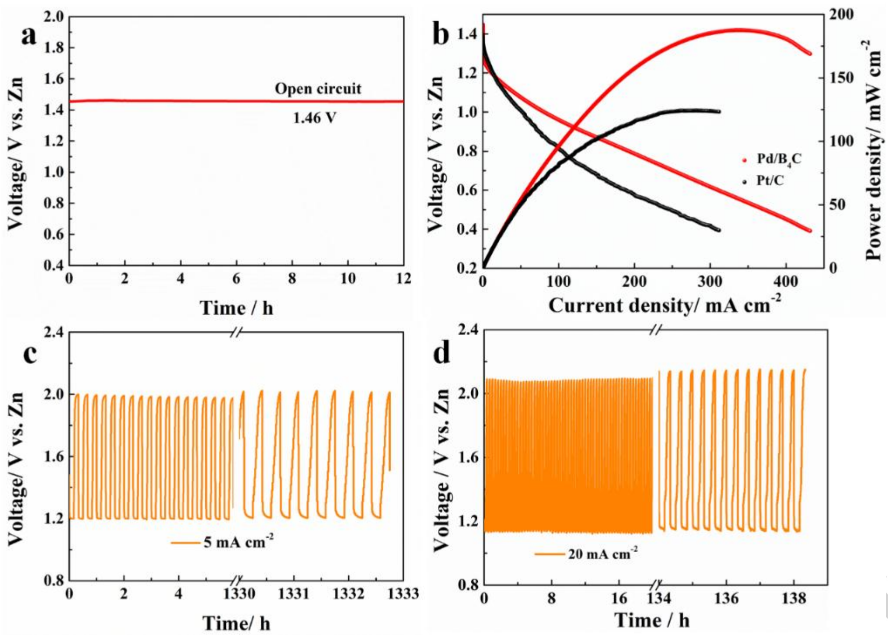


**Fig. 5** (a) The open-circuit plot with of Pd/$B_4C$ as air cathodes for Zn-air batteries. (b) Discharge voltage curves and the corresponding power density plots of Pd/$B_4C$ and Pt/C. (c) Cycling curves at the current density of 5 mA $cm^{-2}$ and (d) 20 mA $cm^{-2}$ of Pd/$B_4C$.

The performance of Pd/$B_4C$ catalysts was then evaluated in Zn-air batteries, with 6 M KOH and 0.2 M $Zn(Ac)_2$ as the electrolyte. Meanwhile, Pt/C was also tested under the same operation condition for comparison. The assembled Zn-air battery with Pd/$B_4C$ as the cathode could work stably with a high open-circuit potential (OCP) of 1.46 V (Fig. 5a) for 12 h, which is higher than that of the Pt/C catalyst (Fig. S17). Besides, the discharge polarization curves and the corresponding power density of Pd/$B_4C$ and Pt/C catalysts are shown in Fig. 5b, from which we could obviously find that the Pd/$B_4C$ electrode shows a relatively higher peak power density of 187 mW $cm^{-2}$. Note that this value is larger than those of most reported catalysts

[18,21,45-49]. These behaviours demonstrate that $Pd/B_4C$ possesses not only high stability, but also superior ORR activity in alkaline electrolytes. Furthermore, in view of the self-carrying OER activity, charging and discharging voltage curves were also tested. From Fig. S18, the voltage gap of $Pd/B_4C$ is close to that of $Pt/C+IrO_2$ (Pt/C: $IrO_2$=1:1) at the current density of 50 mA $cm^{-2}$. We further evaluated the stability of $Pd/B_4C$ in alkaline electrolytes by conducting galvanostatic discharge-charge tests. As shown in Fig. 5c, the $Pd/B_4C$ battery could stably cycle as long as 1333 h (4000 cycles) at 5 mA $cm^{-2}$ without any decrease in the discharge voltage. Even at the current density of 20 mA $cm^{-2}$, Zn-air batteries with $Pd/B_4C$ cathodes could still keep a long cycling life up to 138 h (415 cycles), showing only a slight decrease in the discharge voltage by 20 mV. Note that such ultra-long cycling stability is not only superior to that of the batteries with Pt/C cathodes (Figs. S19 and S20), but also much better than those in reported Zn-air batteries (Table S2). The outstanding cyclic stability is mainly attributed to the corrosion-resistant property of $Pd/B_4C$ composites, especially the chemical and electrochemical stability of $B_4C$ substrate in hash alkaline electrolytes. To further verify the stability of $Pd/B_4C$ in the hash electrolyte, we conducted XPS tests of the electrode before cycles and after 10 cycles (Fig. S21). The B1s spectra mainly contains two peaks located at 187.9 eV and 189.3 eV, which can be assigned to B-C bond in $B_4C$ and $BC_2O$, respectively [50]. Fig. S21b clearly shows the slight existence of Pd(Ⅱ) species on the surface of Pd metal [51]. Note that for both B1s and Pd3d, the ratio of the areas of different peaks keep almost no remarkable change before cycles and after 10 cycles, demonstrating the stability of $Pd/B_4C$ in hash alkaline electrolytes.

## 4. Conclusion

In conclusion, we have successfully prepared roust $Pd/B_4C$ composites with controlled particle sizes. The $Pd/B_4C$ composites could serve as an alkaline-tolerant electrocatalyst in Zn-air batteries, enabling excellent cycling stability of over 1333 h. First-principles computations revealed that the synergistic effect between Pd and $B_4C$ is crucial to generate excellent ORR catalytic activity. This work provides a promising substrate, chemically and electrochemically stable $B_4C$, for constructing high activity and stable catalysts, especially in hash alkaline environment.

## Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version at doi: XXX.

## Acknowledgements

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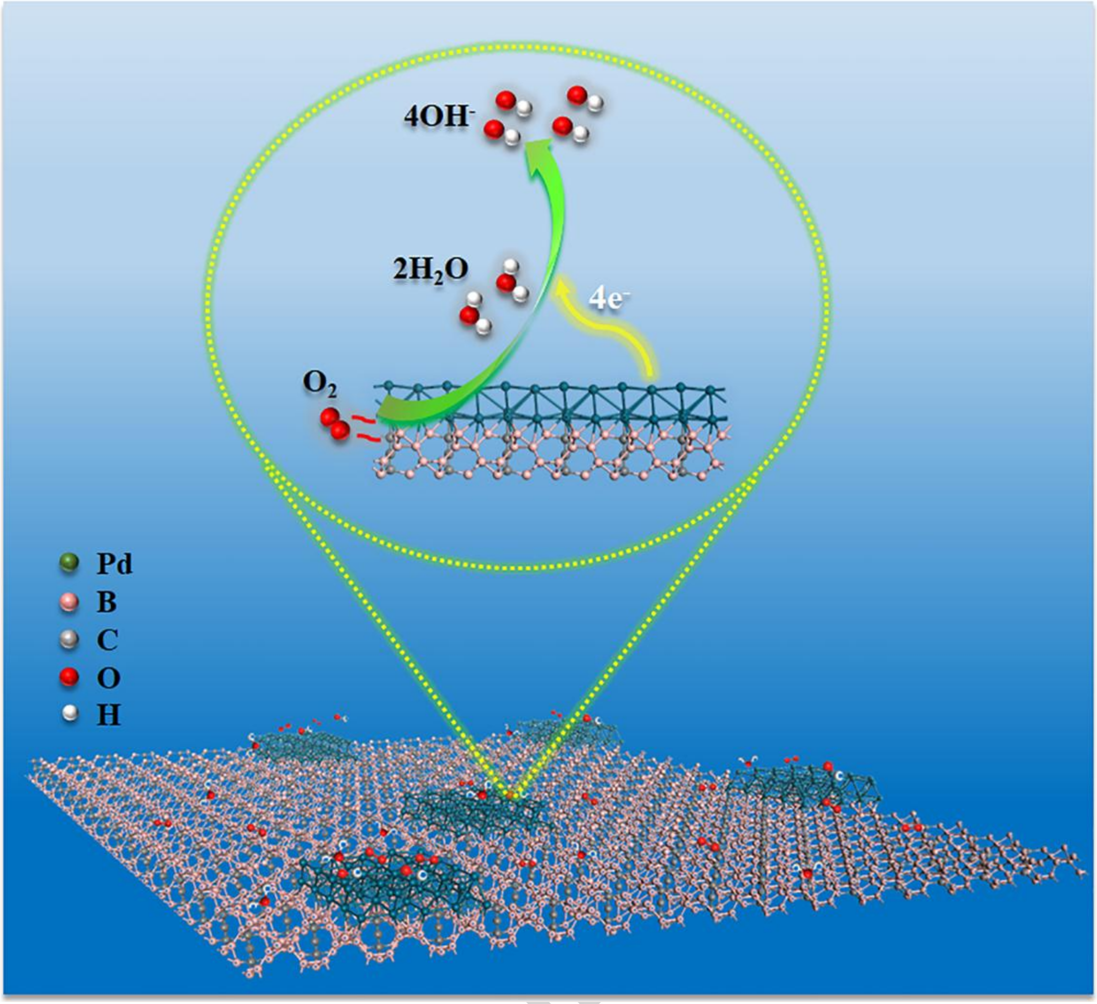
**Research Highlight**

- An electrocatalyst of Pd nanoparticles supported on $B_4C$ was prepared.
- The composite exhibited excellent elctrocatalytic activity and ultralong cycling durability in Zn-air batteries.
- Both experiments and computations reveal the synergistic effect between Pd and $B_4C$ for boosted ORR kinetics.




graphical abstract